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RESEARCH

Sensorineural hearing loss after pediatric cranial radiotherapy: a multicenter analysis of dosimetric and clinical risk factors

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Abstract

Background No multicenter study has validated cochlear dose constraints for hearing preservation in pediatric brain tumor patients in a real-world setting. Although the PENTEC review proposed a 35 Gy mean cochlear dose threshold, supporting evidence from heterogeneous, multicenter pediatric cohorts remains scarce.

Purpose To evaluate dosimetric, therapeutic, and clinical risk factors for sensorineural hearing loss (SNHL) after pediatric cranial radiotherapy in a national multicenter cohort, with a specific focus on validating the clinical relevance of the 35 Gy mean cochlear dose threshold.

Methods We retrospectively analyzed 88 children treated with cranial radiotherapy between 2014 and 2024 across four French pediatric radiotherapy centers participating in the national PediaRT registry. All patients had baseline and at least two post-radiotherapy audiograms graded according to the Chang Ototoxicity Scale, with SNHL defined as Chang grade $\geq$ 1a. Mean (Dmean) and minimum (Dmin) cochlear doses were extracted and analyzed using Kaplan–Meier estimates and Cox proportional hazards models.

Results Over a median follow-up of 37 months, 17 patients (19.3%) developed SNHL. In multivariate analysis, both a mean cochlear dose $\geq$ 35 Gy (HR = 3.90; 95% CI: 1.24–12.23; $p = 0.020$) and cisplatin exposure (HR = 3.85; 95% CI: 1.40–10.64; $p = 0.009$) were independently associated with SNHL. The 5-year cumulative incidence of SNHL was 43.4% for Dmean $\geq$ 35 Gy versus 13.1% for $<$ 35 Gy ($p = 0.0014$), and 51.6% with cisplatin versus 12.4% without ($p < 0.001$). A significant monotonic relationship between cochlear dose and SNHL severity was observed ($p = 0.0011$).

Conclusion This study provides the first national, multicenter validation of the 35 Gy mean cochlear dose threshold in a real-world pediatric population and identifies cisplatin exposure as a major independent risk factor. These findings support strict cochlear dose minimization during treatment planning and systematic long-term audiological surveillance in pediatric cancer survivorship care.

Keywords Brain Neoplasms · Pediatric Radiotherapy · Cochlea · Sensorineural Hearing Loss · Cisplatin

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Introduction

Sensorineural hearing loss (SNHL) is a well-documented late effect of cranial radiation therapy (RT) in children with brain tumors, especially when combined with cisplatin-based chemotherapy. Pediatric survivors are particularly vulnerable to auditory damage, with younger age at treatment and higher cochlear radiation dose being consistently associated with increased risk of SNHL [1–3]. Given the crucial role of hearing in language development and neurocognitive maturation, radiation-induced SNHL can have profound and lasting impacts on academic achievement, social interaction, and quality of life [4–6].

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The cochlea is a radiosensitive structure whose proximity to posterior fossa tumors makes it difficult to spare during treatment. Despite advances in conformal and intensity-modulated RT techniques, cochlear doses often exceed recommended thresholds. The recent Pediatric Normal Tissue Effects in the Clinic (PENTEC) review [7] consolidated evidence from over 450 cochleae, suggesting that the risk of SNHL remains below 5% for mean cochlear doses under 35 Gy, but increases substantially at higher doses. This threshold has since been integrated into several pediatric RT planning strategies.

In our previously published single-center study using data from the French national PediaRT registry [8], we identified cochlear dose and age at RT initiation as the two strongest predictors of SNHL onset in a cohort of 44 children treated between 2014 and 2021. We observed that no patients receiving a mean cochlear dose below 35 Gy developed SNHL, whereas children exceeding this dose had a five-year cumulative incidence above 50%. However, our analysis was limited by sample size and by its monocentric nature, potentially limiting generalizability.

This study extends our previous findings by incorporating data from additional French pediatric RT centers, forming the largest French multicenter cohort. We introduce time-to-event variables to better capture the kinetics of SNHL development and aim to validate previously reported cochlear dose thresholds in a more heterogeneous, real-world population.

Beyond its implications for pediatric radiation oncology, accurately defining and validating cochlear dose constraints has direct consequences for radiotherapy planning in brain tumors and for survivorship guidelines. By confirming a clinically actionable dose threshold in a multicenter, real-world setting, our study provides data that can immediately inform treatment planning decisions aimed at preserving neurocognitive development, academic potential, and quality of life.

Materials and methods

Study design and patient selection

We conducted a retrospective multicenter cohort study based on data from the French national Pediatric RadioTherapy registry (PediaRT). Four institutions participated in the study: Institut de Cancérologie de Lorraine (Nancy), Institut de Cancérologie Strasbourg Europe (Strasbourg), Institut Curie (Paris), and Institut de Cancérologie de l'Ouest (Nantes). Eligible patients were children and adolescents treated with cranial RT for brain tumors between 2014 and 2024, with no prior history of RT or sensorineural hearing loss (SNHL) at baseline audiological evaluation.

Inclusion criteria were: (i) availability of pre-RT audiological assessment; (ii) at least two post-RT audiological evaluations; and (iii) availability of complete dosimetric data for both cochleae. Patients who underwent re-irradiation after initial treatment were excluded.

Treatment characteristics

Radiation therapy data were extracted from the PediaRT registry. Participating centers employed modern conformal techniques, primarily intensity-modulated radiation therapy (IMRT) and volumetric-modulated arc therapy (VMAT). Dosimetric data included total prescribed dose, number of fractions, and mean (D_{mean}), minimum (D_{min}), and maximum (D_{max}, corresponding to $D_{2\%}$) doses to both cochleae and, when available, internal auditory canals.

Treatment protocols varied according to tumor histology and institutional practice, in accordance with ongoing national or international clinical trials. Organs-at-risk (OARs) were contoured according to local guidelines, and contouring practices were left at the discretion of each center, reflecting real-world variability in clinical practice.

All dosimetric parameters were normalized to equivalent doses in 1.8 Gy fractions to allow comparisons across different treatment regimens.

Cisplatin administration, presence of cerebrospinal fluid (CSF) shunting, and history of brain surgery were also recorded as potential covariates impacting hearing outcomes.

Audiological assessment

Baseline audiograms were obtained prior to the initiation of radiotherapy. Follow-up audiological assessments were performed annually or more frequently based on clinical indications. Audiometric data included pure-tone audiometry across conventional frequencies (0.5–8 kHz). Hearing outcomes were graded according to the Chang Ototoxicity Grading Scale (see Table 1), with SNHL defined as a grade ≥ 1a. The Chang ≥ 1a threshold was selected for its sensitivity to early high-frequency hearing loss, which represents the earliest detectable manifestation of radiation-induced ototoxicity and is clinically relevant for pediatric language development. For patients with asymmetric hearing loss, the worse ear was used for analysis.

New variables were introduced to capture the kinetics of hearing loss development, including time from radiation start to first documented SNHL and Chang grade at first detection, in addition to the latest available Chang grade at follow-up.

Table 1 The Chang Ototoxicity Grading Scale

Grade	Hearing threshold
0	≤ 20 dB at 1, 2 and 4 kHz
1a	≥ 40 dB at 6—12 kHz
1b	> 20 and < 40 dB at 4 kHz
2a	≥ 40 dB at ≥ 4 kHz
2b	> 20 and < 40 dB at < 4 kHz
3	≥ 40 dB at ≥ 2 kHz
4	≥ 40 dB at ≥ 1 kHz

Statistical analysis

Descriptive statistics were used to summarize patient, tumor, and treatment characteristics. The cumulative incidence of SNHL was estimated using Kaplan–Meier methods.

Univariate Cox proportional hazards models were used to identify predictive factors for SNHL occurrence, including patient age at RT initiation, tumor characteristics, cisplatin exposure, cochlear dosimetric parameters, and center effects. Significant variables were further assessed in multivariate Cox models with stepwise selection to address potential overfitting due to event number limitations. Mean cochlear dose was predefined as the primary dosimetric variable of interest, in line with the PENTEC review. Analyses of minimum (Dmin) and maximum (Dmax) cochlear doses were exploratory.

Variables reaching statistical significance ($p < 0.05$) in univariate analyses were selected for the multivariate model. Given the limited number of events ($n = 17$), the number of covariates was restricted to two in order to maintain an acceptable events-per-variable ratio (EPV = 8.5). A sensitivity analysis using a clinically significant SNHL threshold (Chang ≥ 2a) was performed to assess the robustness of the primary findings. All dosimetric analyses were conducted at the patient level using the worse ear (i.e., the cochlea receiving the higher dose), consistent with standard Chang grading practice and to avoid the statistical non-independence of bilateral observations.

Comparisons between dosimetric subgroups (e.g., cochlear $D_{mean} < 35$ Gy vs. ≥ 35 Gy) were performed using log-rank tests. A significance level of 0.05 was used for all analyses. All statistical analyses were performed using R version 4.2.2 (2022—10—31).

The authors used OpenAI's ChatGPT to assist in language refinement and editing of the manuscript, without influencing the scientific content.

Results

Patient characteristics

A total of 88 pediatric patients treated with cranial RT across four French centers were included in this study (see Table 2). The cohort comprised 62% males and 38% females, with a median age at RT initiation of 9.5 years (range: 2–17 years). The majority of patients (86%) were older than 5 years at the start of RT, while only 7% were under 3 years of age. The median audiologic follow-up, defined as the time from RT initiation to the last available audiogram, was 37 months (range: 4–124 months; IQR: 27–63 months). Forty-four patients (50%) had follow-up exceeding 3 years, and 24 (27%) exceeding 5 years.

The most frequent tumor type was medulloblastoma (49%), followed by ependymoma (19%), craniopharyngioma (8%), astrocytoma (5%). Nearly all patients (93%) underwent prior neurosurgical intervention, and 45% required cerebrospinal fluid (CSF) shunting. Cisplatin-based chemotherapy was administered to 27% of patients.

In terms of radiation delivery technique, tomotherapy was used in 52% of cases, followed by VMAT (34%), proton therapy (10%), and stereotactic radiotherapy (3%). The mean cochlear dose across the cohort was 28.0 Gy (SD ± 15.9), with 40% of patients receiving a dose ≥ 35 Gy.

Incidence and risk factors of SNHL

Among the 88 children included in this study, 17 (19.3%) developed SNHL during post-treatment follow-up. Analysis of baseline characteristics revealed three factors significantly associated with the occurrence of SNHL (see Table 2). The use of cisplatin-based chemotherapy was strongly associated with increased risk (65% vs. 18% in patients without SNHL, $p < 0.001$), as was a higher minimal ($p = 0.001$) and mean ($p = 0.002$) cochlear dose.

Figure 1 displays the Kaplan–Meier estimate of SNHL-free survival for the entire cohort. The estimated cumulative incidence of SNHL was 4.7% (95% CI: 0.1—9.1%) at 1 year, 6.0% (95% CI: 0.8—11.4%) at 2 years, and 25.6% (95% CI: 11.8—37.3%) at 5 years post-radiation therapy. These results highlight the progressive nature of radiation-induced ototoxicity over time.

Given these findings, both mean cochlear dose and cisplatin exposure were selected for further analysis in time-to-event models.

Table 2 Baseline characteristics by SNHL status

Variable	Overall N = 881	No SNHL N = 711	SNHL N = 171	p-value[2]
Centre				0.3
ICANS	4 (4%)	2 (3%)	2 (12%)	
ICL	44 (50%)	37 (52%)	7 (41%)	
ICO	29 (33%)	22 (31%)	7 (41%)	
Institut Curie	11 (13%)	10 (14%)	1 (6%)	
Gender				0.2
F	33 (38%)	29 (41%)	4 (24%)	
M	55 (62%)	42 (59%)	13 (76%)	
Age at RT initiation (years)				> 0.9
< 3 years	6 (7%)	5 (7%)	1 (6%)	
≥ 3–5 years	6 (7%)	5 (7%)	1 (6%)	
> 5 years	76 (86%)	61 (86%)	15 (88%)	
Tumor type				0.2
Astrocytoma	4 (5%)	3 (3%)	1 (6%)	
Craniopharyngioma	7 (8%)	7 (10%)	0 (0%)	
Ependymoma	17 (19%)	14 (20%)	3 (18%)	
Medulloblastoma	43 (49%)	31 (44%)	12 (70%)	
Other tumor types	17 (19%)	16 (23%)	1 (6%)	
Cisplatin-based chemotherapy	24 (27%)	13 (18%)	11 (65%)	< 0.001
CSF shunt	40 (45%)	29 (41%)	11 (65%)	0.076
Cerebral surgery	82 (93%)	67 (94%)	15 (88%)	0.3
Radiation delivery technique				0.063
VMAT	30 (34%)	28 (39%)	2 (12%)	
Proton therapy	9 (10%)	8 (11%)	1 (6%)	
Stereotactic radiotherapy	3 (4%)	2 (4%)	1 (6%)	
Tomotherapy	46 (52%)	33 (46%)	13 (76%)	
Dmin Cochlea, Gy (Mean ± SD)	25.6 ± 15.5	23.1 ± 16.0	35.9 ± 6.4	0.001
Dmean Cochlea, Gy (Mean ± SD)	28.0 ± 15.9	25.5 ± 16.5	38.2 ± 6.5	0.002
Cochlear dose group				< 0.001
Dmean < 35 Gy	53 (60%)	49 (69%)	4 (24%)	
Dmean ≥ 35 Gy	35 (40%)	22 (31%)	13 (76%)	
Dmax Cochlea, Gy (Mean ± SD)	30.5 ± 17.0	29.4 ± 17.2	35.0 ± 15.9	0.2

[1]n (%)

[2]Fisher's exact test; Pearson's Chi-squared test; Wilcoxon rank sum test

Dose–response relationship between cochlear irradiation and SNHL

Radiation dose to the cochlea was strongly associated with the risk of developing SNHL. Patients who developed SNHL had a significantly higher average dose to the cochlea (38.2 ± 6.5 Gy) compared to those without hearing loss (25.5 ± 16.5 Gy, $p = 0.002$). When applying a dose threshold of 35 Gy, a clear separation between risk groups emerged: 76% of patients with SNHL had received a mean dose ≥ 35 Gy, compared to only 31% in the non-SNHL group ($p < 0.001$) (Fig. 2A).

This association was further confirmed in the time-to-event analysis. Kaplan–Meier curves stratified by the 35 Gy threshold showed a significantly lower probability of remaining SNHL-free in the high-dose group (Fig. 2B). The 5-year cumulative incidence of SNHL reached 43.4% (95% CI: 15.5–62.0%) in patients receiving ≥ 35 Gy versus 13.1% (95% CI: 0.0–25.4%) in those receiving < 35 Gy ($p = 0.0014$, log-rank test).

Impact of cisplatin-based chemotherapy on SNHL

Cisplatin-based chemotherapy was significantly associated with the risk of SNHL in this cohort. While 27% of all patients received cisplatin, the proportion rose to 65% among those who developed SNHL, compared to only 18% in the non-SNHL group ($p < 0.001$) (Table 2).

Figure 3 shows the Kaplan–Meier curves stratified by cisplatin exposure. The probability of remaining

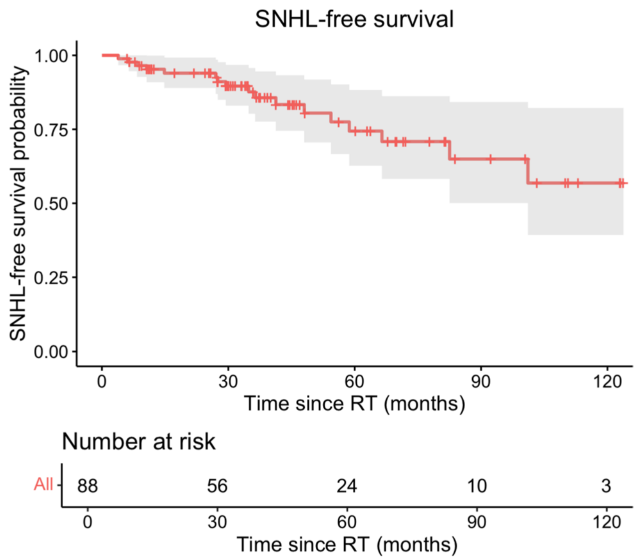


Fig. 1 Probability of remaining free from radiation-induced sensorineural hearing loss (SNHL) over time

SNHL-free was significantly higher in the non-cisplatin group. At 5 years, the cumulative incidence of SNHL reached 12.4% [95% CI: 2.1–21.6%] in patients who did not receive cisplatin, versus 51.6% [95% CI: 16.6–71.9%] in the cisplatin group ($p < 0.001$, log-rank test).

SNHL severity and dose relationship

At the last available audiological follow-up, the severity of sensorineural hearing loss (SNHL) was graded according to the Chang Ototoxicity Scale. Among the 88 patients, 5 children (5.7%) had mild SNHL (grades 1a–1b), while 12 patients (13.6%) experienced moderate to severe hearing loss (grade ≥ 2a). Figure 4A illustrates the distribution of mean cochlear dose according to the final Chang grade. A visual trend was observed toward increasing cochlear dose with worsening SNHL severity. Patients with higher Chang grades tended to have received higher mean doses to the cochlea. This observation was confirmed by a non-parametric Jonckheere–Terpstra test, which demonstrated a significant monotonic trend between cochlear dose and SNHL severity (JT = 971, $p = 0.0011$). These results support a dose–response relationship not only in terms of SNHL occurrence but also in terms of its clinical severity.

When SNHL cases were stratified into mild (grades 1a–1b) versus moderate/severe (grades ≥ 2a), mean cochlear doses tended to be higher in the moderate/severe group (see Fig. 4B). However, the difference was not statistically significant ($p = 0.56$, Wilcoxon rank sum test).

Multivariate analysis

To assess the independent contribution of cochlear dose and cisplatin exposure to the risk of SNHL, a multivariate Cox regression model was performed including both variables

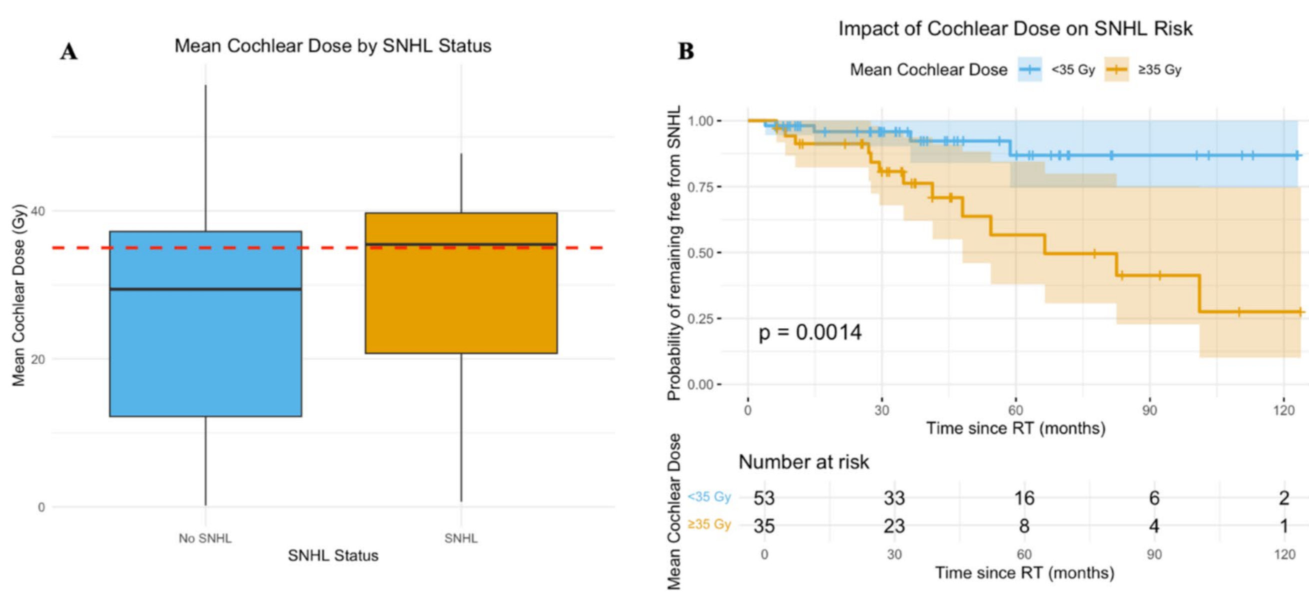
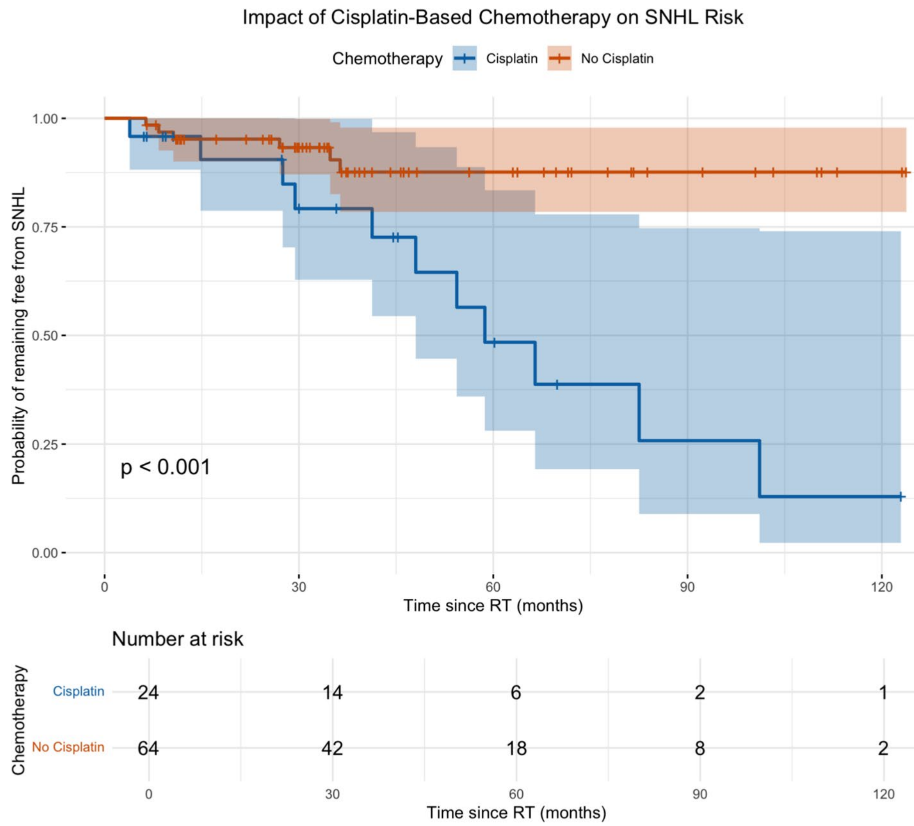


Fig. 2 Association between mean cochlear dose and risk of sensorineural hearing loss (SNHL)*(A) Boxplot of mean cochlear dose in patients with and without SNHL. Patients with SNHL had significantly higher cochlear doses (mean 38.2 Gy vs. 25.5 Gy, p = 0.002). The red dashed line indicates the 35 Gy threshold.(B) Kaplan–Meier curves stratified by mean cochlear dose (< 35 Gy vs. ≥ 35 Gy). The difference was statistically significant (p = 0.0014, log-rank test). Shaded areas represent 95% confidence intervals. Tick marks indicate censoring. Numbers at risk are displayed below the x-axis*

Fig. 3 Impact of cisplatin-based chemotherapy on sensorineural hearing loss (SNHL) risk



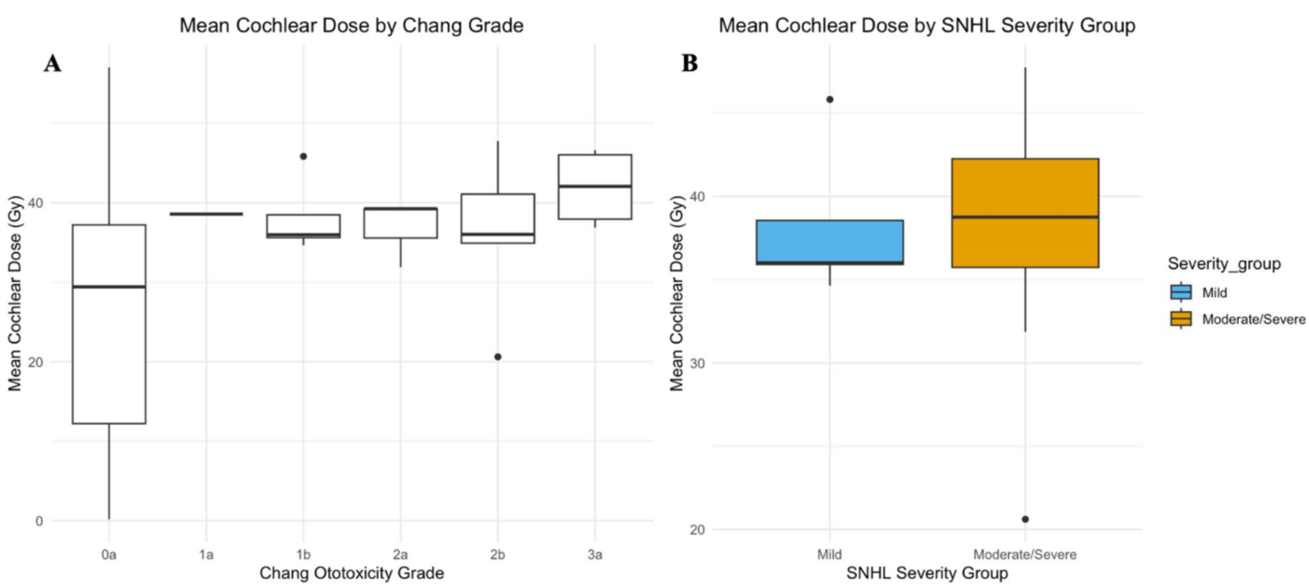


Fig. 4 Relationship between mean cochlear dose and SNHL severity at last follow-up.*(A) Mean cochlear dose according to the final Chang ototoxicity grade in 88 pediatric patients. A monotonic trend toward higher doses with increasing hearing loss severity was observed and confirmed by a Jonckheere–Terpstra test (p = 0.0011). (B) Comparison of mean cochlear dose between patients with mild SNHL (grades 1a–1b) and those with moderate to severe SNHL (grades ≥ 2a). Although cochlear doses tended to be higher in the moderate/severe group, the difference was not statistically significant (p = 0.56, Wilcoxon rank sum test)*

Table 3 Multivariate Cox Model for SNHL Risk

Variable	Hazard Ratio (95% CI)	p-value
Cochlear mean dose ≥ 35 Gy	3.90 (1.24–12.23)	0.020*
Cisplatin-based chemotherapy	3.85 (1.40–10.64)	0.009**

p < 0.05 (*), p < 0.01 (**), p < 0.001 (***)

(see Table 3). After adjustment, both remained significantly associated with SNHL occurrence.

Patients receiving a mean cochlear dose ≥ 35 Gy had a 3.9-fold increased hazard of developing SNHL compared to those receiving < 35 Gy (HR = 3.90; 95% CI: 1.24–12.23; p = 0.020). Similarly, patients treated with cisplatin-based chemotherapy had a 3.85-fold increased risk (HR = 3.85; 95% CI: 1.40–10.64; p = 0.009). The model demonstrated good discriminative performance, with a concordance index of 0.728.

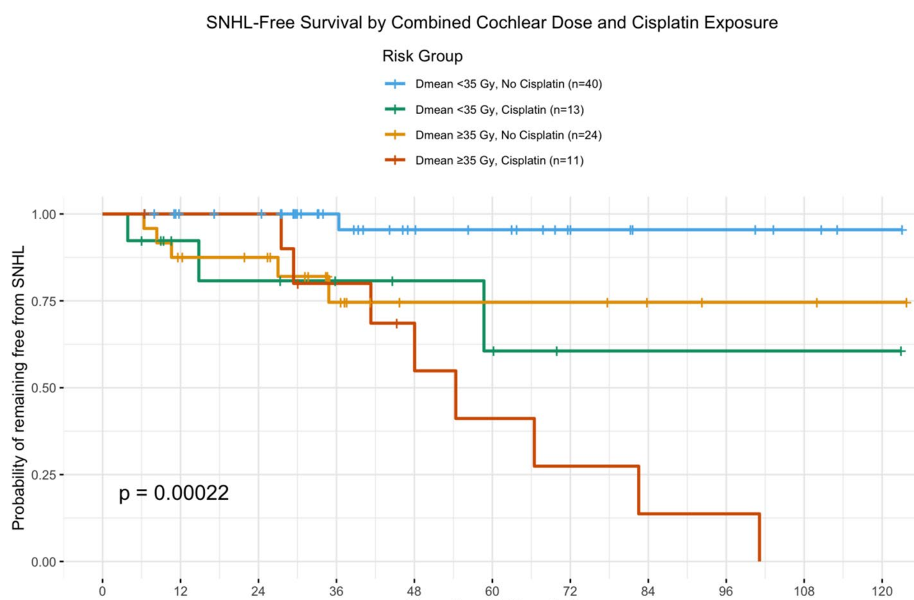
Interaction between cochlear dose and cisplatin exposure

To investigate whether the effects of cochlear dose and cisplatin exposure on SNHL risk were additive or synergistic, a multiplicative interaction term (Dmean ≥ 35 Gy × cisplatin) was added to the Cox regression model. The interaction term was not statistically significant (HR = 0.22; 95% CI: 0.02–2.77; p = 0.24), and the likelihood ratio test comparing models with and without the interaction term confirmed the absence of significant interaction (LR = 1.53, p = 0.22). These results indicate that cochlear irradiation and cisplatin exert additive rather than synergistic effects on SNHL risk. The SNHL rates across combined exposure groups ranged from 2.5% (Dmean < 35 Gy, no cisplatin) to 72.7% (Dmean ≥ 35 Gy with cisplatin), illustrating the cumulative impact of both risk factors (Fig. 5).

Influence of minimum cochlear dose on SNHL risk

Minimum cochlear dose (Dmin) was significantly higher in patients who developed SNHL compared to those who did not (median 37.0 Gy vs. 24.7 Gy, p = 0.001, Wilcoxon rank sum test) (Fig. 6A). Kaplan–Meier analysis stratified by a 20 Gy threshold, selected based on the findings of Cohen-Cutler et al. [9], confirmed a significantly lower probability of remaining SNHL-free in the ≥ 20 Gy group (p = 0.014, log-rank test) (Fig. 6B).

In univariate Cox regression, Dmin was significantly associated with SNHL risk both as a continuous variable (HR = 1.06 per Gy; 95% CI: 1.02–1.11; p = 0.006) and as a



Risk Group	0	12	24	36	48	60	72	84	96	108	120
Dmean <35 Gy, No Cisplatin (n=40)	40	35	34	22	15	13	8	5	5	3	1
Dmean <35 Gy, Cisplatin (n=13)	13	8	7	5	4	3	1	1	1	1	1
Dmean ≥35 Gy, No Cisplatin (n=24)	24	20	18	10	5	5	5	3	2	2	1
Dmean ≥35 Gy, Cisplatin (n=11)	11	10	10	7	5	3	2	1	1	0	0

Fig. 5 SNHL-free survival stratified by combined cochlear dose and cisplatin exposure. *Kaplan–Meier curves showing the probability of remaining free from sensorineural hearing loss (SNHL) according to four risk groups defined by mean cochlear dose (< 35 Gy vs. ≥ 35 Gy) and cisplatin exposure (yes vs. no). The 5-year cumulative incidence of SNHL ranged from 4.5% in the lowest-risk group (Dmean < 35 Gy, no cisplatin; n = 40) to 58.9% in the highest-risk group (Dmean ≥ 35 Gy, cisplatin; n = 11), consistent with an additive effect of cochlear irradiation and cisplatin on hearing loss risk (interaction term p = 0.24). The global log-rank test was statistically significant (p = 0.0002). Tick marks indicate censored observations. Numbers at risk are displayed below the x-axis*

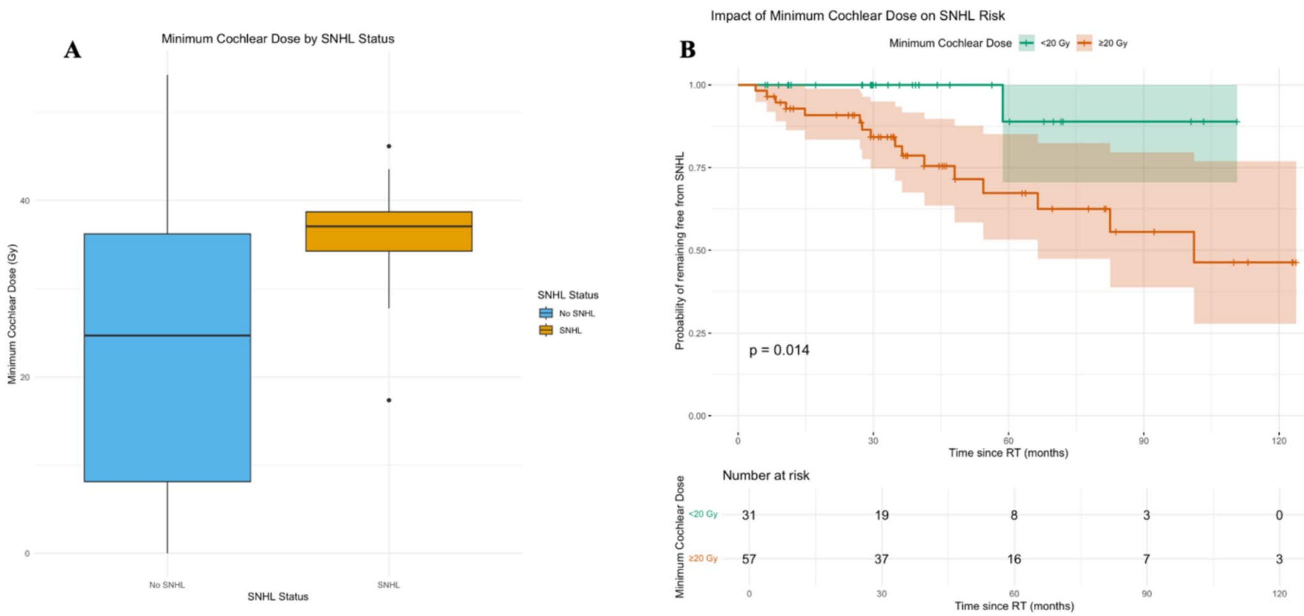


Fig. 6 Association between minimum cochlear dose and sensorineural hearing loss (SNHL) risk.*(A) Boxplot showing the distribution of minimum cochlear dose (Dmin) according to SNHL status. Patients who developed SNHL had significantly higher Dmin values compared to those without hearing loss (median 37.0 Gy vs. 24.7 Gy, p = 0.001, Wilcoxon rank sum test).(B) Kaplan–Meier curves of SNHL-free survival stratified by minimum cochlear dose (< 20 Gy vs. ≥ 20 Gy), a threshold derived from the findings of Cohen-Cutler* et al. *A statistically significant difference was observed between the two groups (p = 0.014, log-rank test). Shaded areas represent 95% confidence intervals. Tick marks indicate censored observations. Numbers at risk are displayed below the x-axis*

categorical variable (≥ 20 Gy vs. < 20 Gy; HR = 8.29; 95% CI: 1.10–62.6; p = 0.04). In multivariate analysis adjusting for cisplatin exposure, Dmin showed a trend toward significance (HR = 6.77; 95% CI: 0.89–51.5; p = 0.065), while cisplatin remained independently associated with SNHL (HR = 4.35; 95% CI: 1.60–11.8; p = 0.004) (Table 4).

Discussion

Cochlear dose and the 35 Gy threshold

This multicenter retrospective study confirmed that SNHL remains a frequent and clinically significant late effect in children receiving cranial RT, despite advances in conformal techniques. Among the 88 patients included, 19.3% developed SNHL during follow-up, with mean cochlear dose ≥ 35 Gy identified as an independent risk factor (HR = 3.90; 95% CI: 1.24–12.23; p = 0.020).

These results are consistent with the landmark findings of Hua et al. [2], who first identified mean cochlear dose as a key predictor of SNHL and recommended limiting doses below 35 Gy. The PENTEC review [7] subsequently consolidated evidence from multiple pediatric cohorts, demonstrating a sharp increase in SNHL risk beyond this threshold. Our study strengthens these observations in a multicenter, real-world setting, confirming that a mean cochlear dose below 35 Gy remains a clinically actionable constraint for hearing preservation, consistent with other recent observational data. Keilty et al. [10] similarly reported a dose–response relationship between cochlear dose and hearing loss severity, proposing an even lower dose goal of ≤ 30 Gy, with a further

Table 4 Summary of survival analyses involving minimum cochlear dose (Dmin)

Analysis	Main result	p-value
Wilcoxon test Dmin vs. SNHL	Higher Dmin in SNHL patients	0.001**
Kaplan–Meier Dmin group (< 20 vs. ≥ 20 Gy)	Significant	0.014*
Univariate Cox regression (continuous Dmin)	HR = 1.06 (1.02–1.11)	0.006**
Univariate Cox regression (binary Dmin ≥ 20 Gy)	HR = 8.29 (1.10–62.6)	0.04*
Multivariate Cox regression (Dmin + Cisplatin)	Dmin HR = 6.77 (p = 0.065), Cisplatin HR = 4.35 (p = 0.004)	-

p < 0.05 (*), p < 0.01 (**), p < 0.001 (***)

reduction to 20–25 Gy for patients receiving platinum-based chemotherapy. Although our data validate the 35 Gy threshold from the PENTEC review, the lower thresholds proposed by Keilty et al. deserve further evaluation in prospective multicenter studies.

Beyond the occurrence of SNHL, we observed a significant monotonic trend between cochlear dose and hearing loss severity, as measured by the Chang ototoxicity scale and confirmed by a Jonckheere–Terpstra test ($p = 0.0011$). Although the comparison between mild and moderate/severe SNHL did not reach statistical significance ($p = 0.56$), likely due to the limited number of events, this finding supports a dose–severity relationship that extends beyond a simple binary outcome and reinforces the importance of minimizing cochlear dose whenever feasible.

Combined ototoxicity of radiation and cisplatin

Cisplatin-based chemotherapy was another major independent risk factor, with a 3.85-fold increased hazard of SNHL after adjustment for cochlear dose (HR = 3.85; 95% CI: 1.40–10.64; $p = 0.009$). Notably, the 5-year cumulative incidence of SNHL reached 51.6% in the cisplatin group despite the use of modern RT techniques (IMRT, VMAT, proton therapy), highlighting the substantial ototoxic burden of combined treatment.

To determine whether these two risk factors act synergistically or additively, we tested the statistical interaction between mean cochlear dose (≥ 35 Gy vs. < 35 Gy) and cisplatin exposure in the Cox model. The interaction term was not significant ($p = 0.24$), indicating that cochlear irradiation and cisplatin exert additive rather than synergistic effects on SNHL risk. This finding is consistent with the results of Keilty et al. [10], who similarly reported additive effects of cochlear dose and platinum exposure. As illustrated in Fig. 5, the 5-year cumulative incidence of SNHL ranged from 4.5% in patients with low cochlear dose and no cisplatin to 58.9% in those exposed to both risk factors, demonstrating the cumulative impact of combined exposure.

Role of minimum cochlear dose

The impact of minimum cochlear dose (Dmin) on SNHL risk was explored given recent findings suggesting its potential predictive value. In our cohort, Dmin was significantly associated with SNHL in univariate analysis, both as a continuous and categorical variable. However, after adjustment for cisplatin exposure, Dmin showed only a trend toward significance ($p = 0.065$), suggesting that while Dmin carries predictive value, consistent with the findings of Cohen-Cutler et al. [9], mean cochlear dose remains the more robust dosimetric predictor for clinical use.

Clinical implications for survivorship care

These findings carry several immediate clinical implications. First, maintaining a mean cochlear dose below 35 Gy should remain a priority during treatment planning for pediatric brain tumors, particularly when cisplatin-based chemotherapy is anticipated. Second, treatment-related hearing loss should be regarded as a modifiable long-term outcome rather than an unavoidable late effect: incorporating cochlear dose awareness and chemotherapy exposure into survivorship planning may allow pediatric teams to tailor follow-up intensity and initiate earlier supportive interventions.

Systematic and long-term audiological surveillance is essential, particularly in high-risk patients. Early detection of SNHL enables timely intervention, including hearing aids and rehabilitative support, to mitigate downstream impacts on language acquisition, schooling, and psychosocial development. In this context, parental counseling plays a critical role: informing families about the importance of repeated audiometric testing can improve adherence to surveillance protocols and facilitate early acceptance of hearing aids when indicated, thereby supporting children's development during and after cancer treatment. Parents should be reassured that early identification of hearing loss enables effective interventions, including modern hearing aids and educational support, that can substantially mitigate the impact of ototoxicity on their child's everyday life, language development, and academic achievement.

Regarding otoprotective strategies, sodium thiosulfate (STS) has emerged as the most promising agent. The phase 3 ACCL0431 trial [11, 12] demonstrated that delayed STS administration reduced the cumulative incidence of cisplatin-induced hearing loss by approximately 50% in children with various cancer types, leading to FDA approval for pediatric patients with localized, non-metastatic solid tumors. However, concerns regarding potential interference with cisplatin efficacy currently limit its use in patients with disseminated disease. The ongoing ACNS2031 trial (ClinicalTrials.gov NCT05382338) is specifically evaluating STS in children with medulloblastoma, a population at particularly high risk for combined radiation- and cisplatin-induced ototoxicity. Results from this trial will be critical to define whether STS can be safely integrated into treatment protocols for this specific population.

Strengths and limitations

The strengths of this study include its multicenter design involving four French centers, the use of the national prospective PediaRT registry, and the rigorous collection of both dosimetric and audiological data. The diversity of tumor types, treatment techniques, and institutional practices in our cohort mirrors the heterogeneity encountered in pediatric neuro-oncology, enhancing the generalizability of our findings.

However, several limitations must be acknowledged. First, the number of SNHL events ($n = 17$) remains modest, which constrains the power of multivariate analyses. With 17 events and 2 covariates, our events-per-variable ratio of 8.5 is below the commonly cited threshold of 10, though this is generally considered acceptable when the concordance index indicates good discriminative performance (C-index = 0.73). To assess the robustness of our findings, a sensitivity analysis using a clinically significant SNHL threshold (Chang ≥ 2a) was performed: both cochlear dose and cisplatin exposure remained associated with increased risk, though the former did not reach statistical significance ($p = 0.07$) due to reduced power while cisplatin exposure remained highly significant (HR = 6.21; $p = 0.007$). Kaplan–Meier analysis confirmed significantly higher incidence of Chang ≥ 2a hearing loss in patients receiving cisplatin (log-rank $p = 0.0002$) and in those with Dmean ≥ 35 Gy (log-rank $p = 0.008$) (Supplementary Table S1).

Second, cisplatin exposure was recorded as a binary variable without information on cumulative dose, number of cycles, or timing relative to radiotherapy. This simplification may underestimate the dose-dependent nature of cisplatin ototoxicity. Similarly, carboplatin exposure was not evaluated, despite its recognized, though less frequent, ototoxic potential [13].

Third, audiometric assessments were limited to conventional frequencies (0.5–8 kHz), whereas the Chang scale defines grade 1a at 6–12 kHz. Since extended high-frequency audiometry was not systematically performed at all centers, the incidence of early-stage SNHL may have been underestimated. This limitation underscores the importance of including high-frequency testing (≥ 12 kHz) in future pediatric audiological protocols, as these frequencies are particularly relevant for speech development in young children.

Fourth, hearing outcomes were graded using the Chang ototoxicity scale rather than the newer SIOP Boston scale. The Chang scale was chosen for its sensitivity to high-frequency hearing loss and for consistency with our prior single-center publication. However, we acknowledge that the SIOP scale has become the preferred standard in platinum-related ototoxicity trials and may have facilitated broader interstudy comparisons.

Fifth, the median audiologic follow-up of 37 months, although 27% of patients had follow-up exceeding 5 years, may be insufficient to capture all late-onset SNHL events. Among the 17 patients who developed SNHL, the median time to diagnosis was 34.8 months (range: 4–101 months); 8 were diagnosed after 3 years, and 3 beyond 5 years. Additionally, 35 censored patients had follow-up shorter than 3 years, suggesting that the true cumulative incidence of SNHL may be higher than reported, and long-term follow-up studies are warranted.

Finally, tumor type was not included in the multivariate model, as cochlear dose distributions differed significantly across tumor types ($p < 0.0001$), with medulloblastoma patients receiving higher cochlear doses and more frequent cisplatin. However, when added to the Cox model, tumor type was not independently associated with SNHL ($p = 0.58$), confirming that its effect is mediated through cochlear dose and chemotherapy exposure rather than representing an independent risk factor.

Conclusion

Sensorineural hearing loss remains a significant late complication in children treated with cranial radiotherapy, particularly in those receiving cisplatin-based chemotherapy or higher cochlear radiation doses. This multicenter study confirms the clinical relevance of a 35 Gy mean cochlear dose threshold and highlights the critical role of systematic, long-term audiological surveillance. Early detection and management of hearing impairment should be prioritized to minimize neurocognitive and quality-of-life consequences in childhood cancer survivors.

Author's contributions Conceptualization: WG, VBC;
Methodology: WG, VBC;
Data curation: WG;
Formal analysis: WG;
Investigation: WG, LO, EJ, CD, MJ;
Writing – original draft: WG;
Writing – review & editing: WG, LO, EJ, CD, MJ, VBC;
Supervision: VBC.

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Data availability The datasets generated and analyzed during the current study are anonymized and not publicly available due to privacy and ethical restrictions but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate This study was conducted in accordance with the Declaration of Helsinki. It was based on retrospective analysis of data from the French national Pediatric Radiotherapy registry (PediaRT), a health data warehouse ("Entrepôt de Données de Santé", EDS) hosted by the Institut de Cancérologie de Lorraine and containing dosimetric data from pediatric radiotherapy treatments. The PediaRT registry complies with the French National Commission for Data Protection (CNIL) reference methodology MS-002 (registration number 2231699, conformity date: November 2, 2023). According to French regulations, retrospective studies based on data from a compliant EDS do not require formal approval by an institutional review board or individual informed consent from participants.

Conflicts of interest The authors declare no competing interests.

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